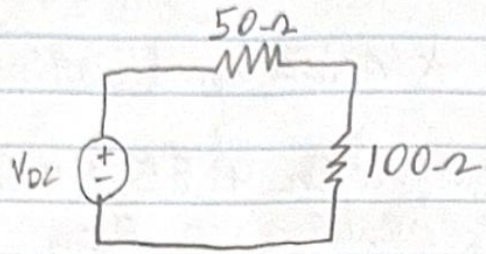
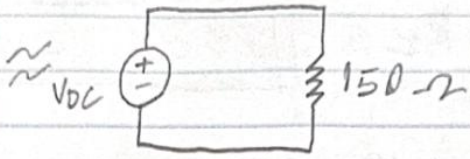
ECE 1000 HW 1
Answer Key

1.

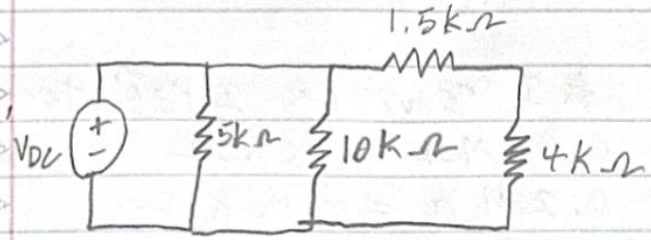


* series combination

$\Rightarrow R_{eq} = 50\Omega + 100\Omega = \boxed{150\Omega}$

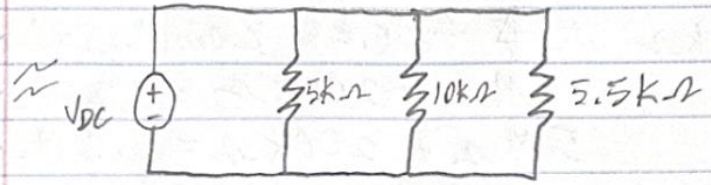


2.



* series combination

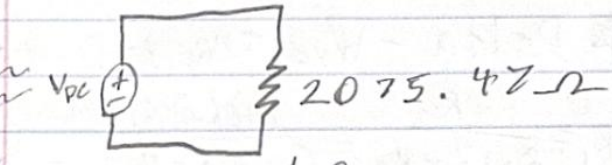
$\Rightarrow 1.5k\Omega + 4k\Omega = 5.5k\Omega$



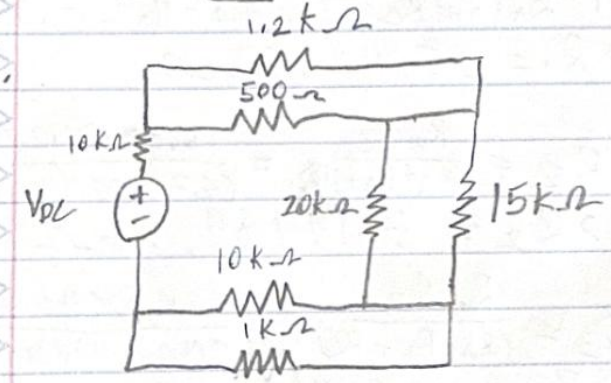
* parallel combination

$\Rightarrow \frac{1}{\left(\frac{1}{5k\Omega}\right)+\left(\frac{1}{10k\Omega}\right)+\left(\frac{1}{5.5k\Omega}\right)} = R_{eq}$

$\Rightarrow R_{eq} = \boxed{2075.47\Omega}$



3.

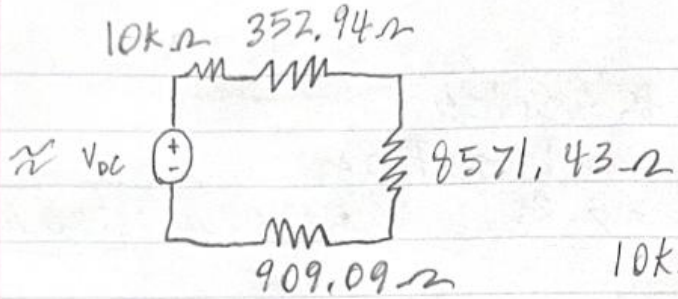


* parallel combinations

$\Rightarrow \frac{1}{\left(\frac{1}{1.2k\Omega}\right)+\left(\frac{1}{500\Omega}\right)} = 352.94\Omega$

$\Rightarrow \frac{1}{\left(\frac{1}{20k\Omega}\right)+\left(\frac{1}{15k\Omega}\right)} = 8571.43\Omega$

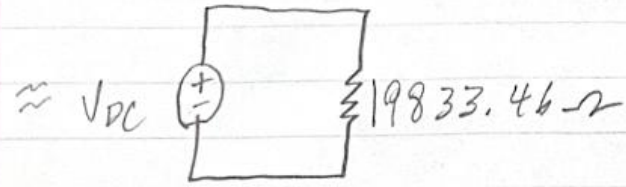
$\Rightarrow \frac{1}{\left(\frac{1}{10k\Omega}\right)+\left(\frac{1}{1k\Omega}\right)} = 909.09\Omega$



★ series combination

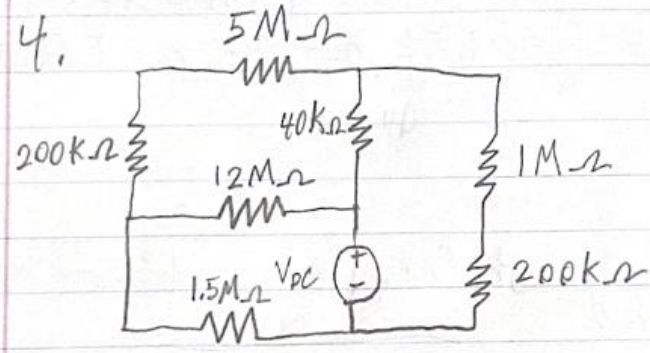
$\Rightarrow 352.94\Omega + 8571.43\Omega$

$10k\Omega + 909.09\Omega = R_{eq}$



$\Rightarrow R_{eq} = \boxed{19833.46\Omega}$

4.



★ I rewrote 2 resistors

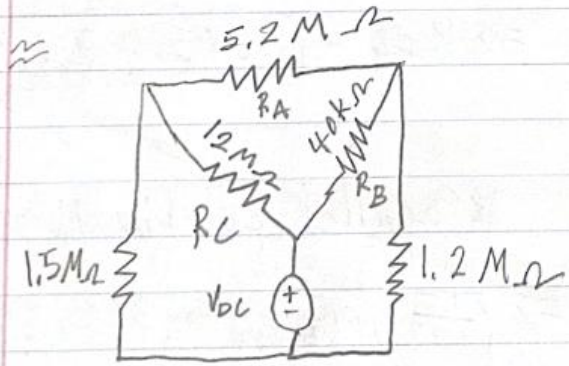
$0.04M\Omega = 40k\Omega$

$0.2M\Omega = 200k\Omega$

★ series combinations

$1M\Omega + 200k\Omega = 1.2M\Omega$

$5M\Omega + 200k\Omega = 5.2M\Omega$



★ Delta - Wye transform

$\Rightarrow R_1 = \frac{R_B R_C}{R_A + R_B + R_C} = \frac{(40k\Omega)(12M\Omega)}{(5.2M\Omega + 40k\Omega + 12M\Omega)}$

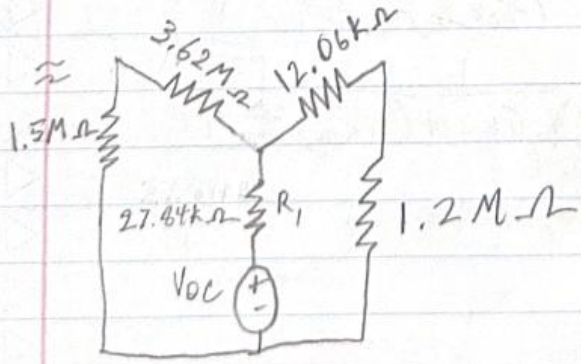
$\Rightarrow R_1 = 27842.23\Omega$

$\Rightarrow R_2 = \frac{R_A R_C}{(R_A + R_B + R_C)} = \frac{(5.2M\Omega)(12M\Omega)}{(5.2M\Omega + 40k\Omega + 12M\Omega)}$

$\Rightarrow R_2 = 3619489.56\Omega$

$\Rightarrow R_3 = \frac{R_A R_B}{(R_A + R_B + R_C)} = \frac{(5.2M\Omega)(40k\Omega)}{(5.2M\Omega + 40k\Omega + 12M\Omega)}$

$\Rightarrow R_3 = 12064.97\Omega$



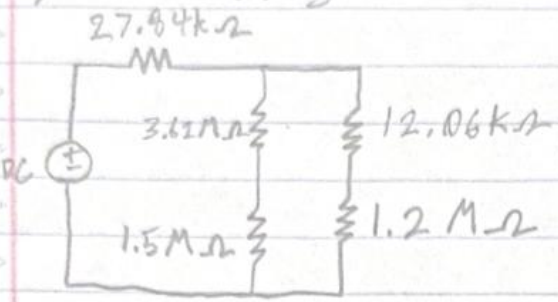
★ simplifying Resistors

$R_1 = 27.84k\Omega$

$R_2 = 3.62M\Omega$

$R_3 = 12.06k\Omega$

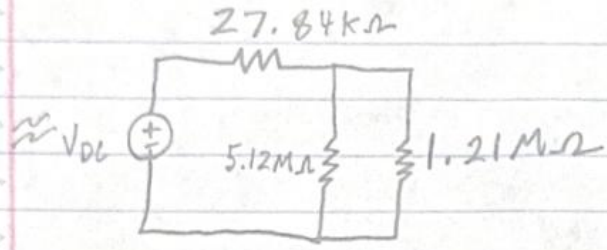
* Redrawing for clarity



* Series combinations

$\Rightarrow 3.62M\Omega + 1.5M\Omega = 5.12M\Omega$

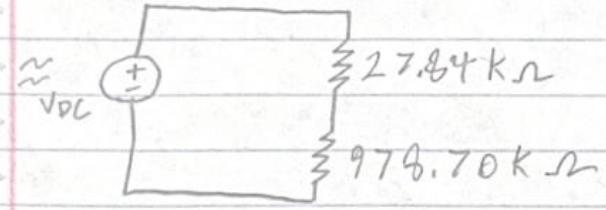
$\Rightarrow 1.2M\Omega + 12.06k\Omega = 1.21M\Omega$



* Parallel combination

$\Rightarrow \frac{1}{\left(\frac{1}{5.12M\Omega}\right)+\left(\frac{1}{1.21M\Omega}\right)} = 978704.58\Omega$

$= 978.70k\Omega$

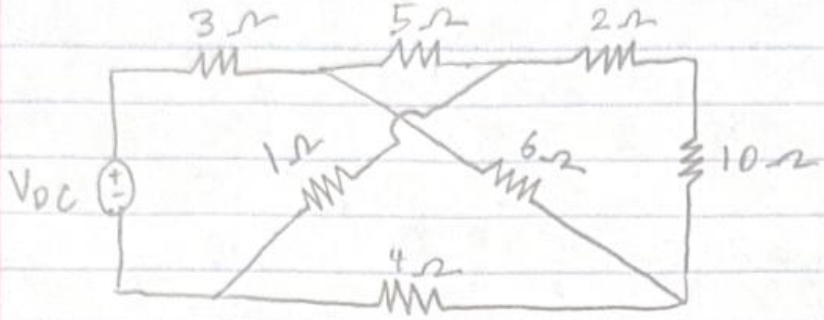


* Series combinations

$\Rightarrow 27.84k\Omega + 978.70k\Omega = R_{eq}$

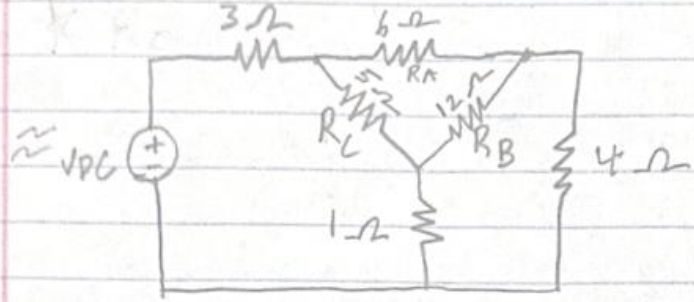
$\Rightarrow R_{eq} = \boxed{1.01M\Omega}$

5.



★ Series combination

$\Rightarrow 2\Omega + 10\Omega = 12\Omega$



★ Delta-Wye transform

$\Rightarrow R_1 = \frac{R_B R_C}{R_A + R_B + R_C} = \frac{(12\Omega)(5\Omega)}{(6\Omega + 5\Omega + 12\Omega)}$

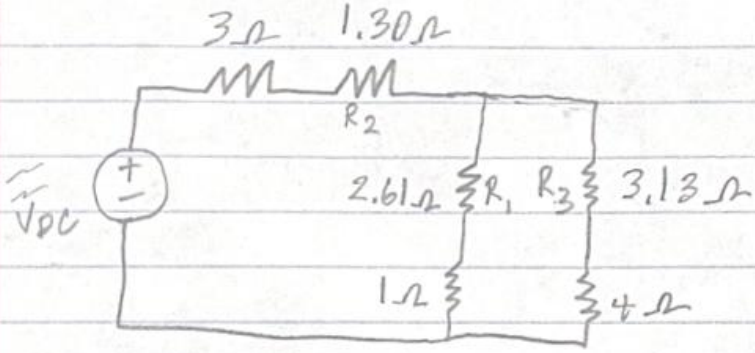
$\Rightarrow R_1 = 2.61\ \Omega$

$\Rightarrow R_2 = \frac{R_A R_C}{R_A + R_B + R_C} = \frac{(6\Omega)(5\Omega)}{(6\Omega + 5\Omega + 12\Omega)}$

$\Rightarrow R_2 = 1.30\ \Omega$

$\Rightarrow R_3 = \frac{R_A R_B}{R_A + R_B + R_C} = \frac{(6\Omega)(12\Omega)}{(6\Omega + 5\Omega + 12\Omega)}$

$\Rightarrow R_3 = 3.13\ \Omega$

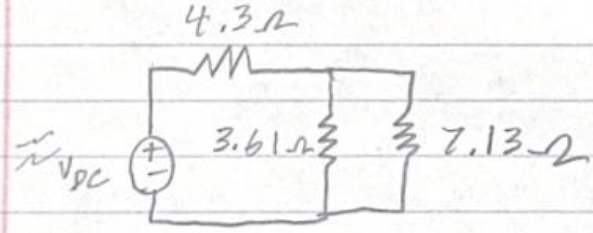


★ Series combinations

$\Rightarrow 3\Omega + 1.30\Omega = 4.3\Omega$

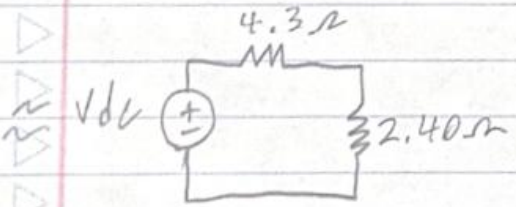
$\Rightarrow 2.61\Omega + 1\Omega = 3.61\Omega$

$\Rightarrow 3.13\Omega + 4\Omega = 7.13\Omega$



★ Parallel combination

$\Rightarrow \frac{1}{\left(\frac{1}{3.61\Omega}\right) + \left(\frac{1}{7.13\Omega}\right)} = 2.40\ \Omega$



★ Series combination

$\Rightarrow 4.3\Omega + 2.4\Omega = R_{eq}$

$\Rightarrow R_{eq} = \boxed{6.7\ \Omega}$